

M. MURALIDHARAN

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PROFESSIONAL SUMMARY

B.Tech Computer Science Engineering graduate (2026) from VIT Chennai with demonstrated expertise in full-stack development, ML-integrated systems, and applied research. Lead author at IEEE CVMI 2025, presenting research on computer vision algorithms for urban safety analysis. Developed a deployable Multi-Agent Reinforcement Learning system achieving a 31% reduction in carbon emissions across a supply chain simulation. Experienced across the complete stack — from React and FastAPI frontends/backends to cloud deployment on AWS — with real-world exposure through an industry internship at Ramco Cements. Available to join immediately.

ACHIEVEMENTS & PUBLICATIONS

IEEE CVMI 2025 – Lead Author (Oct 12–13, 2025): Published and presented research on object tracking algorithms (CSRT, FairMOT, DeepSORT, ByteTrack) for urban intersection risk analysis, demonstrating expertise in computer vision and applied software research.

TECHNICAL SKILLS

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js, Next.js, Tailwind CSS, Streamlit

Backend: Node.js, Express.js, Flask, FastAPI, REST APIs

Mobile: Android (Java)

Databases: MySQL, MongoDB

Cloud & Tools: AWS, Firebase, Git, GitHub

ML / AI: Python, Reinforcement Learning, Qiskit, Facebook Prophet, HDF5, PCA, Random Forest, SVM

Data Viz: Power BI, Tableau

Languages: Python, Java, JavaScript, C++

WORK EXPERIENCE

The Ramco Cements Limited | IT Intern | On-Site, Chennai

Jun 2025 – Jul 2025

- Developed a full-stack application integrating ML-based sales prediction algorithms with a web frontend and database backend, demonstrating end-to-end software development capability.
- Built and deployed interactive Power BI dashboards connected to live data sources, tracking sales volume, regional revenue, and customer demand KPIs.
- Designed an interactive logistics dashboard in Tableau using real-time Excel data, visualizing dispatch frequency, delivery timelines, and efficiency metrics for operational decision-making.
- Collaborated with cross-functional stakeholders to gather requirements, iterate on deliverables, and present data-driven solutions to management.

PROJECTS

Object Tracking for Urban Intersection Risk Analysis — IEEE CVMI 2025 (Lead Author)

Python, PyTorch, OpenCV, YOLOv8, CSRT, DeepSORT, FairMOT, ByteTrack

- Conducted a comparative study of four multi-object tracking algorithms (CSRT, FairMOT, DeepSORT, ByteTrack) for urban intersection risk analysis, building a modular YOLOv8-based detection pipeline to standardize input across trackers for fair comparison.
- Benchmarked trackers across three complementary datasets (MOT17, KITTI, Cityscapes) using MOTA, MOTP, ID switches, and FPS, finding ByteTrack delivered the best accuracy/real-time trade-off (up to 80% MOTA at 32.5 FPS) while FairMOT was most stable in high-pedestrian-density scenes.
- Performed qualitative failure-case and confusion-matrix analysis across all four trackers, translating tracking performance into practical guidance for pedestrian-safety monitoring and smart-transportation deployment.
- Published and presented the work as lead author at the 2025 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI), Oct 12–13, 2025.

MARL Supply Chain Optimization

Python, Reinforcement Learning, FastAPI, Streamlit, Constrained MDP, Graph Attention Networks

- Developed a Multi-Agent RL system (MAPPO + Graph Attention Networks) modeling a 12-node South Indian logistics network where independent agents learn optimized policies under disruption scenarios.
- Achieved a 31% reduction in carbon emissions while maintaining service levels through constrained reward shaping — showcasing algorithmic problem-solving with measurable business impact.
- Built a deployment-ready system with a FastAPI backend and Streamlit dashboard for real-time policy visualization and performance comparison.

Sales Prophet – AI Sales Forecasting System

Python, Flask, Facebook Prophet, MySQL, MongoDB, REST API

- Built a full-stack AI application with a Flask backend, integrating Facebook Prophet for time-series sales forecasting — demonstrating backend API design and database integration skills.
- Used MySQL for structured transactional data and MongoDB for unstructured forecast outputs, reflecting real-world polyglot persistence patterns.
- Exposed forecast results via RESTful APIs and visualized trends through interactive dashboards, connecting backend logic to a user-facing frontend.

EDUCATION

B.Tech – Computer Science Engineering | CGPA: 7.71

Vellore Institute of Technology, Chennai | 2022 – 2026 (Graduated)

Languages: English • Tamil • Telugu